

Software Requirements Specification (SRS)

Inventory Management System

Introduction

The Inventory Management System (IMS) is to simplify and automate inventory-related operations within an organization. The system enables businesses to efficiently manage products, monitor stock levels, record sales and purchases, and generate reports. It minimizes manual work, improves inventory accuracy, and supports better business decisions.

Scope

The Inventory Management System is designed for small and medium-sized businesses. It allows authorized users to:

- Manage product information
- Add and update inventory records
- Record purchases and sales
- Monitor stock availability
- Generate inventory reports
- Maintain supplier information

The system aims to reduce inventory errors, save time, and improve operational efficiency.

Objectives

The objectives of the system are:

- Automate inventory management processes.
- Reduce paperwork and manual calculations.
- Improve inventory accuracy.
- Provide real-time stock information.
- Generate useful reports for management.
- Enhance data security through user authentication.

Overall Description

Product Perspective

The Inventory Management System is a standalone web-based application that stores inventory information in a centralized database. Users can access the system through a web browser after logging in.

Product Features

The major features include:

- User Login
- Dashboard
- Product Management
- Category Management
- Supplier Management
- Purchase Management
- Sales Management
- Inventory Tracking
- Low Stock Alert
- Report Generation

User Classes

There are two main types of users:

Administrator

- Manage users
- Add, edit, and delete products
- Manage suppliers
- Generate reports
- Monitor inventory

Staff

- Record purchases
- Record sales
- View inventory
- Search products

Functional Requirements

The system shall provide the following functions:

FR-1 User Authentication

- Users must log in using a valid username and password.
- Unauthorized users cannot access the system.

FR-2 Product Management

The administrator can:

- Add products
- Edit product information
- Delete products
- Search products

FR-3 Supplier Management

The administrator can:

- Add suppliers
- Update supplier information
- Delete suppliers

FR-4 Inventory Management

The system shall:

- Update stock automatically after purchase.
- Reduce stock automatically after sales.
- Display current stock quantity.

FR-5 Purchase Management

Users can:

- Record new purchases
- View purchase history
- Edit purchase records

FR-6 Sales Management

Users can:

- Record sales

- Generate invoices
- View sales history

FR-7 Report Generation

The system shall generate:

- Daily sales report
 - Monthly sales report
 - Inventory report
 - Low stock report
-

Non-Functional Requirements

Performance

- The system should respond within **3 seconds** for normal operations.
- It should support multiple users simultaneously.

Security

- Passwords must be encrypted.
- Only authorized users can access sensitive information.
- User authentication is required.

Reliability

- The system should maintain accurate inventory data.
- Data loss should be prevented through regular backups.

Availability

- The system should be available 24/7 except during scheduled maintenance.

Usability

- The interface should be user-friendly.
- Navigation should be simple and intuitive.

Maintainability

- The software should allow future updates and modifications without affecting existing functionality.
-

External Interface Requirements

User Interface

The interface should contain:

- Login Page
- Dashboard
- Product Management Page
- Supplier Management Page
- Purchase Page
- Sales Page
- Reports Page

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Dual-Core Processor
- Internet Connection

Software Requirements

- Windows/Linux
 - Google Chrome or Mozilla Firefox
 - MySQL Database
 - PHP/Python/Java (depending on implementation)
-

Use Case Description

Use Case: Login

Actor: User

Precondition

- User must have a registered account.

Main Flow

1. User enters username and password.
2. System verifies credentials.
3. User is redirected to the dashboard.

Postcondition

- User successfully logs into the system.
-

Use Case: Add Product

Actor: Administrator

Precondition

- Administrator is logged in.

Main Flow

1. Select "Add Product."
2. Enter product details.
3. Click Save.
4. System stores the information.

Postcondition

- New product is added to inventory.

Database Design

Product Table

Field	Type
Product_ID	Integer
Product_Name	Varchar
Category	Varchar
Quantity	Integer
Price	Decimal
Supplier_ID	Integer

Supplier Table

Field	Type
Supplier_ID	Integer
Supplier_Name	Varchar
Contact	Varchar
Address	Varchar

8. Assumptions and Constraints

Assumptions

- Users have basic computer knowledge.
- Internet connectivity is available.
- The database server functions correctly.

Constraints

- Only authorized users can modify inventory.
 - The system depends on database availability.
 - Power or network failures may temporarily interrupt service.
-

9. Conclusion

The Inventory Management System provides an efficient solution for managing inventory operations. It automates stock management, reduces manual errors, improves inventory accuracy, and generates useful reports for decision-making. The system enhances business productivity while ensuring secure and reliable data management.